

Team Worksheet — Lab 16

Fishbone, 5 Whys and Cause Prioritisation — In Depth

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-16-cause-prioritisation.xlsx`.

3. What we produced

Deliverable: A fishbone diagram, 5 Whys chains and a prioritised list of testable candidate causes.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Recap brainstorming facilitation in full: divergence before convergence, no criticism during generation, and how to stop the highest-paid opinion from dominating.

Step 2. Recap the fishbone structure: the effect at the head, and 5M+E bones — Manpower, Method, Machine, Material, Measurement, Environment.

Step 3. Write your capstone problem statement precisely at the head of the fishbone. A vague effect produces vague causes.

Step 4. Facilitate the fishbone with your team, generating at least 25 candidate causes across all six bones. Thin bones signal blind spots, not absence of causes.

Step 5. Recap the 5 Whys and its discipline: continue until the answer is a process or system failure, never a person to blame.

Step 6. Apply 5 Whys to your top five fishbone causes. Note where a single chain branches — real causal chains are often trees, not lines.

Step 7. Recap multi-voting and apply it to reduce your cause list to the ten most plausible candidates.

Step 8. Build a cause-and-effect matrix: score each candidate cause against each CTQ, weighted by CTQ importance, to produce a ranked cause list.

Step 9. Filter the ranked list for TESTABILITY — for each cause, can you state a hypothesis and obtain data? An untestable cause cannot be proven in Analyze.

Step 10. Convert your top five testable causes into formal hypothesis statements ready for Lab 17.

Step 11. Note how you would coach a Green Belt team through this session, and the two facilitation mistakes you would watch for.

7. Self-check

Your fishbone carries 25+ causes across all six bones, 5 Whys chains terminate at process rather than person, and you have five testable causes written as formal hypotheses.

Complete? ☐ Yes ☐ Not yet

Certified Lean Six Sigma Black Belt (CLSSBB) Training · TGS-2024051900 · Version v4 · © 2026 Tertiary Infotech Academy Pte Ltd